Wheeler's Delayed Choice with a Single Atom (Truscott et al.) Explained by a Finite-Speed ransactional Handshake The TauTimp option $V = 10^9 c$, the wave–particle duality ellipse, and a consistent causal account of the experiment Viorel-Mihai Popescu — TauTimp / Onduscular Theory working notes 27 September 2026

Abstract

In 2015 Manning, Khakimov, Dall and Truscott performed Wheeler's delayed-choice gedanken experiment with single ultracold metastable helium atoms in a matter-wave Mach–Zehnder interferometer built from laser (Bragg) gratings. The decision to apply the second beam splitter was taken by a quantum random-number generator *after* the atom had passed the first beam splitter. The atom showed interference when the second beam splitter was present and which-path statistics when it was absent, exactly as quantum mechanics predicts. The naive reading — that a later choice fixes whether the atom "had travelled" as a wave or as a particle — suggests retrocausality.

This note gives an explanatory mechanism for the result within the TauTimp framework. The mechanism is a transactional handshake that runs forward in the time of a preferred frame at a finite speed $V = 10^9 c$ (V-TI). It rests on three points. (i) The Bragg pulses are coherent, stimulated, unitary interactions: they leave no record, so no transaction forms before detection. (ii) The only non-unitary event is absorption at the detector, roughly 0.4 s after the choice. At that moment the handshake has to span only the extent of the atomic wave packet, a few centimetres, which takes about 10^{-19} s at speed V. (iii) The outcome probabilities are fixed by the offer wave as it actually is at absorption, so they already include whichever second-beam-splitter configuration was chosen. No influence has to go backward in time.

We connect the mechanism to the wave–particle duality ellipse of Khatiwada and Qian (Phys. Rev. Research 7, 033033, 2025). The random choice becomes a random selection of the effective coherence at the absorber, $\gamma_{\text{eff}} \in \{0,1\}$. V-TI adds one factor, the handshake-completion function $f(\xi)$, to the ellipse. For the Truscott parameters $\xi \sim 10^{18}$, so $f = 1$ and V-TI reproduces the observed data exactly. We state openly that the experiment is *explained* by V-TI but does not *test* it, and we identify the regime in which V-TI's prediction would differ from standard quantum mechanics.

1. Introduction

Wheeler's delayed-choice thought experiment (1978) sharpens Bohr's complementarity. A quantum enters an interferometer. Only afterwards does the experimenter decide whether to recombine the two paths, which reveals interference ("wave"), or to leave them separate, which reveals which path was taken ("particle"). If the quantum "decided" its nature at the entrance, a later choice should not be able to change the statistics. Yet it always does. Wheeler's conclusion was that "no elementary phenomenon is a phenomenon until it is a registered phenomenon". Talk of what the quantum "was doing" inside the interferometer before detection has no meaning.

Optical versions were carried out by Jacques et al. (2007) and, with the choice spacelike-separated over 144 km, by Ma et al. (2013). The experiment of Truscott's group at the Australian National University (Manning et al., *Nature Physics* **11**, 539, 2015) was the first with a **single massive particle**. It is therefore the cleanest test of the claim that the effect depends on light being "wavy".

The aim of this note is not to question the quantum-mechanical prediction; the data agree with it. The aim is to provide a **mechanism**: a physical account of *when* and *where* the outcome is fixed that is (a) consistent with the data, (b) free of backward-in-time causation, and (c) quantitatively tied to the TauTimp constant $V = 10^9 c$ used elsewhere in this corpus (see *TauTimp_TIQM_Superluminal_Handshake.md*).

2. The Truscott experiment

2.1 Setup

- **Particle.** Helium-4 atoms in the metastable 2^3S_1 state (He*), released from a Bose–Einstein condensate. On average about one atom per run takes part in the interferometer, so the statistics are single-atom statistics.
- **Beam splitters and mirror.** Pulses of counter-propagating laser beams near the 1083 nm He* transition form a moving optical lattice. Through Bragg diffraction this lattice transfers momentum $2\hbar k$ to the atom coherently. A $\pi/2$ **pulse** acts as a 50/50 beam splitter, putting the atom into the superposition $|p_0\rangle$ and $|p_0 + 2\hbar k\rangle$. A π **pulse** acts as a mirror and swaps the two momenta.
- **Sequence.** $\pi/2$ (BS_1) at $t = 0$, then π (M) at $t = T$, then the optional $\pi/2$ (BS_2) at $t = 2T$. T is on the millisecond scale.
- **Choice.** A quantum random-number generator (QRNG) based on vacuum-noise measurement decides whether BS_2 is applied. The random bit is generated *after* the atom has passed BS_1, so the atom's entry into the interferometer cannot "know" the configuration.
- **Detection.** The atom falls about 0.85 m under gravity (about 0.42 s) onto a microchannel-plate delay-line detector (MCP-DLD). This detector records single-atom arrival positions and times. The two output momentum states arrive at different places and times, so the output port is read out directly.

2.2 Result

- **BS_2 present:** the output-port populations oscillate with the relative phase φ, showing interference fringes of high visibility ("wave").
- **BS_2 absent:** each port receives half the atoms regardless of φ, with no fringes ("particle", since the port identifies the path).

The late random choice therefore determined which of the two complementary behaviours appeared, in full agreement with quantum mechanics.

2.3 Characteristic scales (computed)

Quantity	Expression	Value
Atomic mass (He-4)	m	6.646×10^{-27} kg
Bragg recoil velocity (counter-propagating, upper bound)	$v_r = 2h/(\lambda m)$	0.184 m/s
Arm separation at $t = T$	$\Delta x = v_r T$	18 μm ($T = 0.1$ ms) … 184 μm ($T = 1$ ms)
Fall height / time to detector	$L,\ t_A = \sqrt{2L/g}$	0.848 m, 0.416 s
Spread of wave packet at detector (BS_2 absent)	$\ell_{\mathrm{OW}} \approx v_r t_A$	≈ 7.7 cm

Note. If the Bragg beams are not exactly counter-propagating, v_r is multiplied by $\sin(\theta/2)$. None of the conclusions below depend on this factor. They hold for any value within several orders of magnitude.

3. Standard quantum description

3.1 State evolution

Write $|a\rangle \equiv |p_0\rangle$ and $|b\rangle \equiv |p_0 + 2\hbar k\rangle$. With ideal pulses:

$$\mathrm{BS}_1: \quad |a\rangle \ \rightarrow \ \frac{1}{\sqrt{2}}(|a\rangle + i\,|b\rangle),$$

$$\mathrm{M}: \quad |a\rangle \leftrightarrow i\,|b\rangle, \qquad |\psi(2T^-)\rangle = \frac{1}{\sqrt{2}}\left(i\,|b\rangle - e^{i\varphi}|a\rangle\right),$$

where φ collects the laser phases and the propagation phases of the two arms.

BS_2 present. A second $\pi/2$ pulse mixes $|a\rangle$ and $|b\rangle$ once more. The port probabilities are

$$P_a = \cos^2\frac{\varphi}{2}, \qquad P_b = \sin^2\frac{\varphi}{2}, \qquad \mathcal{V} = \frac{P_{\max} - P_{\min}}{P_{\max} + P_{\min}} = 1.$$

BS_2 absent. The state is not mixed again:

$$P_a = P_b = 1/2 \quad \text{for all } \varphi, \qquad \mathcal{V} = 0.$$

The random choice is a choice between two different **final unitaries** applied before the single measurement. Nothing in the formalism refers to "what the atom was" between BS_1 and BS_2.

3.2 The duality ellipse (Khatiwada & Qian 2025) applied to the experiment

Khatiwada and Qian write the general two-path state as

$$|\Psi\rangle = c_1\,|1\rangle|M_1\rangle + c_2\,|2\rangle|M_2\rangle, \qquad \gamma = |\langle M_1|M_2\rangle|,$$

where $|M_{1,2}\rangle$ contain every remaining degree of freedom (internal states, environment) and γ is the degree of coherence. They show that the visibility $\mathcal{V} = 2|c_1 c_2|\gamma$ and the predictability $D = ||c_1|^2 - |c_2|^2|$ satisfy the exact **duality ellipse (DE)**

$$\frac{\mathcal{V}^2}{\gamma^2} + D^2 = 1, \qquad \eta = 1 - \gamma \ \text{(ellipticity)},$$

which reduces to $\mathcal{V}^2 + D^2 = 1$ for full coherence. With imperfect alignment (α) and imperfect source coherence it generalises to $\mathcal{V}^2/(\gamma^2 T^2 \alpha^2) + D^2 = 1$ (their eq. 11). Its form is unchanged; only the ellipticity grows.

Application to delayed choice. Let $|\chi_1\rangle, |\chi_2\rangle$ be the detector modes that branches 1 and 2 finally reach. The fringe visibility is then

$$\mathcal{V} = 2|c_1 c_2|\ \gamma_{\mathrm{env}}\ \gamma_{\mathrm{eff}}, \qquad \gamma_{\mathrm{eff}} \equiv |\langle\chi_1|\chi_2\rangle_{\text{port basis}}|,$$

where γ_{eff} measures how far the two branches end up in the *same* set of detector modes.

- BS_2 absent: branch 1 lands in momentum port a and branch 2 in port b. These are orthogonal, so $\gamma_{\mathrm{eff}} = 0$, $\mathcal{V} = 0$, and Englert's distinguishability is $\mathcal{D} = \sqrt{1 - \gamma_{\mathrm{eff}}^2} = 1$.
- BS_2 present: both branches are spread over both ports with complementary phases, so $\gamma_{\mathrm{eff}} = 1$, $\mathcal{V} = 2|c_1 c_2|\gamma_{\mathrm{env}}$, and $\mathcal{D} = 0$.

Because the $\pi/2$ pulse is balanced, $D = 0$ in both cases, and the DE reduces to $\mathcal{V} = \gamma_{\mathrm{env}}\gamma_{\mathrm{eff}}$. **Wheeler's delayed choice is thus a random choice of the ellipticity of the duality ellipse, made just before detection:** $\eta = 1$ (degenerate, pure particle) or $\eta = 1 - \gamma_{\mathrm{env}}$ (near-circle, wave). This reading needs no statement about the atom's past. It is the key to the mechanism below.

4. The TauTimp mechanism: a finite-speed handshake (V-TI)

4.1 Postulates

P1 — Offer wave. The quantum state ψ is a real physical *offer wave* (OW). It evolves unitarily (Schrödinger/Bragg dynamics) at subluminal group velocity and always follows **every** available path. The OW carries the energy and momentum that can be transferred.

P2 — Two kinds of interaction (after Kastner's RTI). *U-interactions* are coherent and stimulated, and leave no record in any other system: the Bragg pulses, free flight, and phase shifts. They change the OW but form no transaction. *NU-interactions* are events in which an absorber can irreversibly take the quantum, such as the ionisation of an MCP channel by the 19.8 eV internal energy of He*. Only these start a handshake.

P3 — Confirmation and handshake at finite speed. When the OW reaches a set of possible absorbers $\{k\}$, each absorber answers with a confirmation wave (CW) of amplitude $\langle k|\psi\rangle^*$. The CWs are coordinated, and one transaction is selected, by a non-local link that spreads **forward in the time of a preferred frame** Σ_{PF} (natural choice: the CMB rest frame) at speed

$$V = 10^9\, c = 2.998 \times 10^{17}\ \text{m/s}.$$

P4 — Born rule. The transaction with absorber k is actualised with probability equal to the OW–CW echo, $P_k = \langle\psi|k\rangle\langle k|\psi\rangle = |\langle k|\psi\rangle|^2$, evaluated **on the OW as it is at the moment of absorption**.

P5 — Completion condition. The transaction is complete when the link has covered the extent ℓ_{OW} of the offer wave across the set of absorbers. This takes

$$\tau_H = \frac{\ell_{\mathrm{OW}}}{V}.$$

If the OW is changed by a U-interaction less than τ_H before absorption, the CWs come partly from the old configuration and partly from the new one. The resulting interference is a mixture, described by a completion function $f(\xi)$ with

$$\xi = \frac{V\,(t_A - t_{\text{last U}})}{\ell_{\mathrm{OW}}}, \qquad f(0) = 0, \quad f(\xi \geq 1) = 1.$$

4.2 The origin of $V = 10^9 c$ in TauTimp

The TauTimp field equation $\frac{R}{r}\Delta H = \frac{1}{c^2}\partial_t^2 H$ has the characteristic speed $v(r) = c\sqrt{R/r}$. With $R = 3.9228 \times 10^{22}$ m this speed equals $10^9 c$ at $r_V = 10^{-18} R = 39.2$ km $\approx 3.5\, r_0$, where $r_0 = 11\,222$ m is the inner boundary. At r_0 itself it is $1.87 \times 10^9 c$. The value $V = 10^9 c$ is therefore the natural order of magnitude of the fastest characteristic speed of the onduscular equation. How this field-theoretic speed becomes the speed of the transactional link is an open point (see §8).

4.3 Modified duality ellipse

Under P5 the visibility gains one factor:

$$\mathcal{V}_{\text{V-TI}} = 2|c_1 c_2|\ \gamma_{\text{env}}\ \alpha\ \gamma_{\text{eff}}\ f(\xi), \qquad \frac{\mathcal{V}^2}{\gamma_{\text{env}}^2\,\alpha^2\,\gamma_{\text{eff}}^2\,f^2(\xi)} + D^2 = 1$$

V-TI keeps the elliptical form found by Khatiwada and Qian. Its only new ingredient is a **timing-dependent ellipticity** $\eta = 1 - \gamma_{\text{env}}\alpha\gamma_{\text{eff}} f(\xi)$.

5. Application to the Truscott experiment 5.1 Timeline of the single atom

Step	Time (lab)	Type	V-TI interpretation
Release from BEC	$t < 0$	—	OW prepared, $\|a\rangle$
BS_1 $(\pi/2)$	0	U	OW split into $\|a\rangle$, $\|b\rangle$; **no decision, no record**
QRNG produces bit	$0 < t_c < 2T$	(classical)	selects the next U-operation
Mirror (π)	T	U	branches swapped
BS_2 applied / not applied	$2T$ (ms)	U / none	final unitary fixed: $\gamma_{\text{eff}} = 1$ or 0
Free fall	$2T \to t_A$	U	OW spreads to $\ell_{\mathrm{OW}} \sim 8$ cm
MCP absorption	$t_A \approx 0.42$ s	**NU**	CWs from both ports; handshake over ℓ_{OW}
Transaction complete	$t_A + \tau_H$	—	$\tau_H \approx 2.6 \times 10^{-19}$ s

5.2 Numbers

Quantity	Value
$\tau_H = \ell_{\mathrm{OW}}/V$ (wave packet at detector, 7.7 cm)	2.6×10^{-19} s $\approx 1.4\,\tau_q$
$2\Delta x/V$ (handshake across the arms, $T = 1$ ms)	1.2×10^{-21} s
L/V (handshake across the whole apparatus, 0.85 m)	2.8×10^{-18} s $\approx 16\,\tau_q$
Margin between last U-operation and absorption	$t_A - 2T \approx 0.41$ s
Completion parameter $\xi = V(t_A - 2T)/\ell_{\mathrm{OW}}$	$\approx 1.6 \times 10^{18}$
Handshake completion $f(\xi)$	1 (exactly, to all measurable precision)

Here $\tau_q = 1.765 \times 10^{-19}$ s is the TauTimp time quantum (light crossing the Bohr radius). The V-length of one quantum is $V\tau_q = 5.3$ cm, the same order as the atomic wave packet at the detector.

5.3 How the mechanism explains the result

1. **Nothing is decided at BS_1.** BS_1 is a U-interaction: stimulated Bragg scattering returns every absorbed lattice photon to the laser mode and leaves no record of the path. By P1 the OW takes both arms. The question "did the atom go one way or both ways?" is not about any physical event, because no transaction exists yet.
2. **The late choice only selects the last unitary.** Whether the QRNG bit is generated before or after the atom enters is irrelevant. The bit decides whether a U-operation (BS_2) acts on the OW at $t = 2T$. In the Khatiwada–Qian language it sets γ_{eff}.
3. **The only "decision" happens at the detector, about 0.4 s later.** At t_A the OW reaches the MCP. If BS_2 was applied, both ports contain a coherent sum of the two arms, and the CW echoes give $P_{a,b} = \cos^2(\varphi/2), \sin^2(\varphi/2)$. If BS_2 was not applied, each port holds one arm only and $P_{a,b} = 1/2$. These are exactly the observed statistics.
4. **The handshake is complete long before anything could change.** The link needs $\tau_H \approx 3 \times 10^{-19}$ s. The configuration was fixed 0.41 s earlier, a margin of $\sim 10^{18}\tau_H$, so $f(\xi) = 1$ and $\mathcal{V}_{\text{V-TI}} = \mathcal{V}_{\text{QM}}$.
5. **No retrocausality.** Every step runs forward in time: forward in lab time for the OW, forward in PF time for the handshake. "Delayed choice" is not a choice about the past. It is a choice about the last unitary *before the only irreversible event*, and that event lies in the future of the choice. Wheeler's statement that "no phenomenon is a phenomenon until it is registered" becomes a physical statement in V-TI: the registration is the NU-event, and the handshake at speed V is what turns a possibility into a phenomenon.

5.4 What the finite speed adds compared with TIQM and RTI

- Compared with **Cramer's TIQM**, V-TI does not need an advanced wave travelling to the past or an atemporal "pseudo-time". The handshake is an ordinary forward process with a definite duration τ_H.
- Compared with **Kastner's RTI**, V-TI keeps the U/NU distinction but places the actualisation *in* spacetime with speed V, not in a pre-spacetime realm. That makes a quantitative prediction possible (§7).
- Compared with **standard QM**, V-TI gives the same predictions wherever $\xi \gg 1$, which includes every delayed-choice experiment carried out so far.

6. Why the mechanism is valid (consistency checks)

Requirement	Status in V-TI	Justification
Reproduces the Truscott data	✓	$f(\xi) = 1$ for $\xi \sim 10^{18}$, so the standard QM result is recovered (§5.2)
Born rule	✓	$P_k = \lvert\langle k\rvert\psi\rangle\rvert^2$ from the OW–CW echo (P4)
Conservation (energy, momentum, one atom = one click)	✓	a single transaction transfers the whole quantum; other branches are cancelled by the link
Single-particle no-signalling	✓	the outcome at one port is not controllable from elsewhere; the QRNG bit acts only through a local U-operation
No retrocausation	✓	all influences run forward in lab or PF time
Other delayed-choice data (Jacques 2007, Ma 2013)	✓	the switching margins (ns to µs) exceed the corresponding τ_H (fs to ps) by many orders of magnitude
Bell-test speed bounds ($\gtrsim 10^4 c$)	✓	$V = 10^9 c$ lies above all bounds published so far
Lorentz invariance of correlations	✗ (by design)	preferred frame Σ_{PF}; testable, see §7
Multipartite no-signalling (Bancal et al. 2012)	✗ in principle	any finite V allows signalling in 4-party configurations; accepted as a falsifiable prediction

A ruled-out simpler variant. One might try to make coherence *require* the two arms to lie within a single V-cell, $\Delta x < V\tau_q = 5.3$ cm. The Truscott experiment satisfies this (µm–mm), but Jacques et al. (2007) saw full single-photon interference in an interferometer with 48 m arms, and entanglement-based erasure has been shown over 144 km (Ma et al. 2013). The criterion is therefore **false**. In V-TI coherence is set by γ_{env} and γ_{eff} (U-dynamics), not by the V-length. The speed V enters only the *completion time* of the NU-handshake. The document uses only this weaker and valid role of V.

7. Predictions and falsifiability

1. **Last-moment switching.** Suppose a U-operation that sets γ_{eff} could be applied within $\Delta t < \ell_{\text{OW}}/V$ before absorption. V-TI then predicts a reduced visibility $\mathcal{V} = \mathcal{V}_{\text{QM}} f(\xi)$, $\xi < 1$. Standard QM predicts no reduction. For the Truscott geometry this window is $\sim 3 \times 10^{-19}$ s, far below any existing switching technology, so the experiment is **compatible** with V-TI but **does not test** it.
2. **Enlarging the window.** τ_H grows linearly with the transverse size of the OW at the absorber. An optical delayed-choice experiment in which a single photon's mode is spread over a large detector array, e.g. $\ell_{\text{OW}} =$

300 km for a satellite downlink, gives $\tau_H = 1$ ps. That is within reach of modern electro-optic switching (sub-ps). A measurable loss of visibility when the switch is fired within $$1 ps before detection would support V-TI. Its absence would push V above $10^9 c$ or rule the model out.

3. **Sidereal modulation.** Since Σ_{PF} is the CMB frame, τ_H in the lab depends on the orientation of ℓ_{OW} relative to the CMB velocity ($\beta = 1.23 \times 10^{-3}$). A test near the window edge should show a modulation with the sidereal day.
4. **Multipartite test.** The four-party Bancal–Gisin configuration at $V = 10^9 c$ would show signalling within V-spacelike windows. This is V-TI's sharpest and riskiest prediction.

8. Limitations and open problems

- **Derivation of V.** At present $V = 10^9 c$ is the characteristic speed of the onduscular equation at $r_V \approx 3.5\, r_0$. It still has to be shown that the transactional link is carried by the TauTimp field H and propagates at $v(r_V)$ (or at $v(r_0) = 1.87 \times 10^9 c$) in laboratory transactions.
- **Form of $f(\xi)$.** Only its limits are fixed. A physical model is needed, for example an Airy-type transition as in the spherical TauTimp branch.
- **Preferred frame.** Giving up Lorentz invariance of the correlation mechanism needs positive experimental evidence to be accepted.
- **Scope.** This note does not use earlier TauTimp hypotheses about energy thresholds of the observing light or tau-neutrino mediation. They are independent of the mechanism above and would need their own experimental support.

9. Conclusions

The single-atom delayed-choice experiment of Truscott and co-workers is explained, without retrocausality, by a finite-speed transactional handshake. The coherent Bragg pulses, including the randomly chosen second beam splitter, are unitary and fix only the form of the offer wave. The outcome is decided solely at the irreversible absorption event on the detector, roughly 0.4 s later, by a handshake that spans the few-centimetre wave packet in about 3×10^{-19} s at $V = 10^9 c$. In the duality-ellipse language of Khatiwada and Qian, the random choice selects the effective coherence $\gamma_{\mathrm{eff}} \in \{0,1\}$ and hence the ellipticity of the ellipse. V-TI adds only a timing factor $f(\xi)$, which equals 1 for this experiment. The mechanism reproduces all delayed-choice data. It is consistent with the published bounds on the speed of quantum influences, and it makes specific falsifiable predictions for sub-picosecond last-moment switching and for multipartite Bell tests.

References

1. A. G. Manning, R. I. Khakimov, R. G. Dall, A. G. Truscott, "Wheeler's delayed-choice gedanken experiment with a single atom," *Nature Physics* **11**, 539–542 (2015).
2. P. Khatiwada, X.-F. Qian, "Wave-particle duality ellipse and application in quantum imaging with undetected photons," *Phys. Rev. Research* **7**, 033033 (2025).
3. J. A. Wheeler, "The 'past' and the 'delayed-choice' double-slit experiment," in *Mathematical Foundations of Quantum Theory*, A. R. Marlow (ed.), Academic Press (1978).
4. V. Jacques et al., "Experimental realization of Wheeler's delayed-choice gedanken experiment," *Science* **315**, 966 (2007).
5. X.-S. Ma et al., "Quantum erasure with causally disconnected choice," *PNAS* **110**, 1221 (2013).
6. B.-G. Englert, "Fringe visibility and which-way information: an inequality," *Phys. Rev. Lett.* **77**, 2154 (1996).
7. D. M. Greenberger, A. Yasin, "Simultaneous wave and particle knowledge in a neutron interferometer," *Phys. Lett. A* **128**, 391 (1988).
8. J. G. Cramer, "The transactional interpretation of quantum mechanics," *Rev. Mod. Phys.* **58**, 647 (1986).
9. R. E. Kastner, "The Relativistic Transactional Interpretation and Spacetime Emergence," arXiv:2103.11245 (2021); *The Transactional Interpretation of Quantum Mechanics: A Relativistic Treatment*, CUP (2022).
10. J.-D. Bancal et al., "Quantum non-locality based on finite-speed causal influences leads to superluminal signalling," *Nature Physics* **8**, 867 (2012).
11. D. Salart et al., "Testing the speed of 'spooky action at a distance'," *Nature* **454**, 861 (2008).
12. V.-M. Popescu, *Transactional Quantum Mechanics and a Finite Superluminal Handshake* (TauTimp working note, 2026).

Cramer
28 Sep. 2026